

Can simple statistics call a UFC fight?

Three decades of UFC results, scraped from Sherdog

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1 Introduction

Sherdog.com has recorded every UFC event since the promotion's first show in November 1993. Every event is attributed with fights, fighters and details of these exchanges in impressive detail. The UFC is also considered the pinnacle of the fighting world stage and so this public data hosted and recorded by Sherdog.com welcomes interesting analysis on the world's greatest fighting stage. For example, **how much of a fight's outcome is already visible in the two fighters' records beforehand?** is one of the topics of discussion in this report.

I crawled Sherdog's UFC subgraph with Scrapy. Wikipedia was also scraped for additional attendance stats. Both the crawl and scraped included raw data as well as fighter portrait jpegs. All data was validated and cleaned through a four stage item pipeline, and stored as three linked CSVs. The crawl reached every one of the 1134 UFC events listed in Sherdog's sitemaps, plus one more found from the recursive capability of a crawler. Because the spider operates on a deep graph traversal search, going from events, to fights, fighters and back to events or fights, no data escaped the spider. A duplicate UFC URL base case was used to end the recursion. A nice demonstration of real crawling vs scraping.

The pipeline discarded 299 Ultimate Fighter episodes and non UFC promotions ensuring only genuine historic UFC events remained. This was done to stay consistent with the research question.

Following are my findings summarised. Decisions rose from 14% of fights to 51%. Submissions fell from 37% to 16%. Within submissions the technique mix narrowed around chokes. Home advantage is worth one percentage point, and a layoff costs nothing until it passes a year. Finally, a simple rule built from three pre-fight signals was built and back tested against the historic outcomes. The rule achieved a successful prediction rate of **58.7%**. A modest edge, but an advantage nonetheless against a 50% coin flip.

2 Implementation

Items

Four scrapy.Item subclasses cover the domain. EventItem holds one card, FightItem one bout and FighterItem one biography. EventAttendanceItem is a fourth item and unrelated to Sherdog. Values like event date and height needing normalisation carry both a raw and a parsed field, and every record carries a source_url and scraped_at field.

Spiders

EventsSpider is the main crawler. parse_sitemap reads four XML event sitemaps, strips namespaces and filters to /events/UFC- URLs. parse_event yields the event plus one FightItem per bout. The undercard event is a separate table to the main event and is handled accordingly. parse_fighter yields the biography and portrait URL and then follows the UFC rows of that fighter's record back to further events and

opponents. This makes the run a crawl and not a scrape. A second spider, `AttendanceSpider`, is used to target Wikipedia’s “List of UFC events” and is parsed into one `EventAttendanceItem` per row.

Pipeline

Four stages run in sequence. `CleaningPipeline` first *validates*. A non-UFC or Ultimate Fighter event, or an attendance row with an unparseable date, raises `DropItem` before parsing. What’s left is normalised, with dates parsed, `method_raw` split by regex, height/weight converted to metric, attendance figures stripped of commas and the “N/a” placeholder Wikipedia uses for unreported gates.

Scrapy’s `ImagesPipeline` downloads each fighter’s portrait. `ImageMetadataPipeline` is a small custom stage that actually *processes* the file, opening it with `Pillow` to read real pixel dimensions and size rather than just recording where it landed.

The `CsvStoragePipeline` only opens the CSVs the running spider needs so one spider cannot truncate another’s output. Items are routed to CSVs by their item type, and field names are read from the `Item` classes, ensuring the CSV schema will not drift from its item definitions.

Settings. By design the crawler is polite. `ROBOTSTXT_OBEY` is true, the user agent names the project with a contact address, and each domain gets one concurrent request with a randomised 1.5 s delay, tuned by `AutoThrottle` at target concurrency 1.0. `HTTPCACHE_ENABLED` allowed pipeline changes to be run against a cached copy of the site. This proved very important when refactoring had to be made during the development process.

3 Crawl process

The crawl is a graph traversal. It seeds from four sitemaps, expands to event pages, and follows both fighters from each event to their profiles. A fighter’s profile exposes that fighter’s UFC record, linking back to further events and opponents. A fighter/event bipartite graph is explored from many directions at once.

The spider terminates because the UFC subgraph is finite and every request is constrained to `/events/UFC-` URLs or a fighter reachable from one. It also does not duplicate work. Scrapy’s duplicate filter contained 54,514 redundant requests, and `fight.csv` contains zero duplicate `fight_id` values. A fight ID had to be created from `{event_id}-{bout_order}`, as `Sherdog` had none of its own. The full crawl, including 3,245 portrait downloads, ran once live over roughly 7.5 hours and replayed from cache thereafter.

Only one non-UFC card listed two bouts both labelled “Match 3”, producing a duplicate `fight_id`. That is an error on `Sherdog`’s side, resolved by the exclusion rule.

4 Data collected

The cleaned corpus is three linked CSVs, namely `events.csv` (836 rows, one UFC event), `fight.csv` (8,989, one bout) and `fighters.csv` (3,256, one fighter).

Coverage is complete. `Sherdog`’s four sitemaps list 1,134 distinct `/events/UFC-` URLs, and the crawl fetched all 1,134. It also fetched one further UFC event that appears in no sitemap, reached only by following a fighter’s record. That is a demonstration that the graph traversal finds pages a sitemap walk would miss. Of those 1,135 pages, the validation stage discarded 299, comprising 293 Ultimate Fighter episodes and 6 events from unrelated promotions whose names collide with the UFC prefix. That left the 836 genuine UFC events, of which 820 carry results. The 16 events that carried no results were future cards at the time of the crawl. A span of fights from 1993-11-12 to 2026-07-25 remained.

Two more datasets sit alongside the three already mentioned. Of the crawled fighters, 3,245 of 3,256 have a portrait. The total size is 44 MB, 3,050 files. The `attendance.csv` holds 708 rows of which only 560 rows actually record an attendance number.

5 Results

Every fighter-level feature (`win_rate_entering`, `finish_rate_entering`, `days_since_prior`, `age_at_fight`) is built with `groupby(...).shift().expanding()`, so a fighter's value entering fight n uses only fights $1 \dots n - 1$. If a fighter's record would include the fight being predicted the analysis would be meaningless.

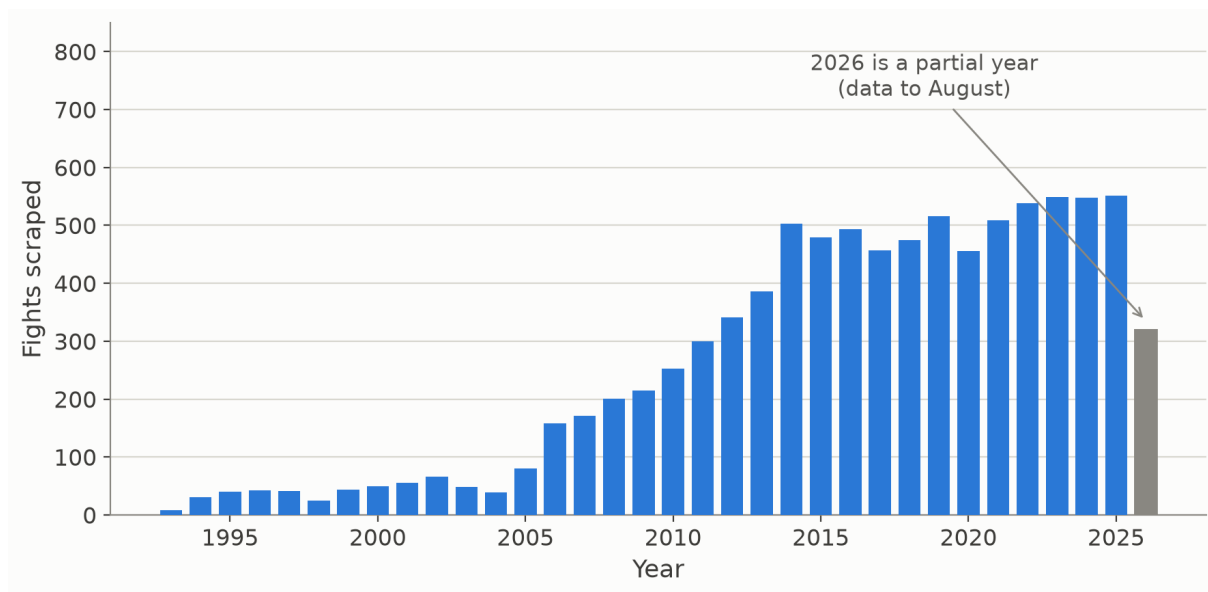


Figure 1: **The UFC grew the quickest from the early 2000s to the mid 2010s.** The UFC has grown considerably over the decades with its sharpest climb from 2005 to 2015.

5.1 How has the way fights end shifted?

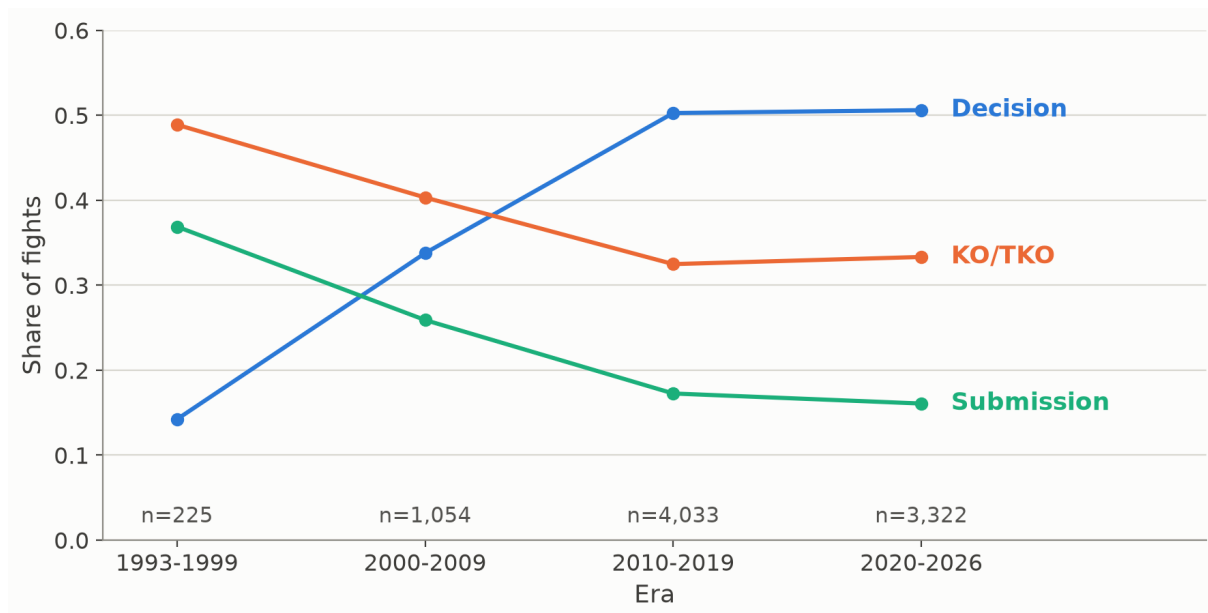


Figure 2: **Decisions have taken over the UFC.** Share of fights ending by each method across four eras. The n beneath each era is the number of fights it rests on.

The trend is decisive. Decisions rose from 14.2% of fights in 1993–1999 to 50.6% in 2020–2026. Submissions fell from 36.9% to 16.1% and KO/TKOs declined gently, from 48.9% to 33.3%. A modern UFC fight is more likely to reach the judges than to be finished.

A possible explanation is that rules and regulations have matured in ways that favour survival. The

promotion has shifted from a battle of risk taking and submission, to a game of sprawling, pinning, time wastage and ground dominance.

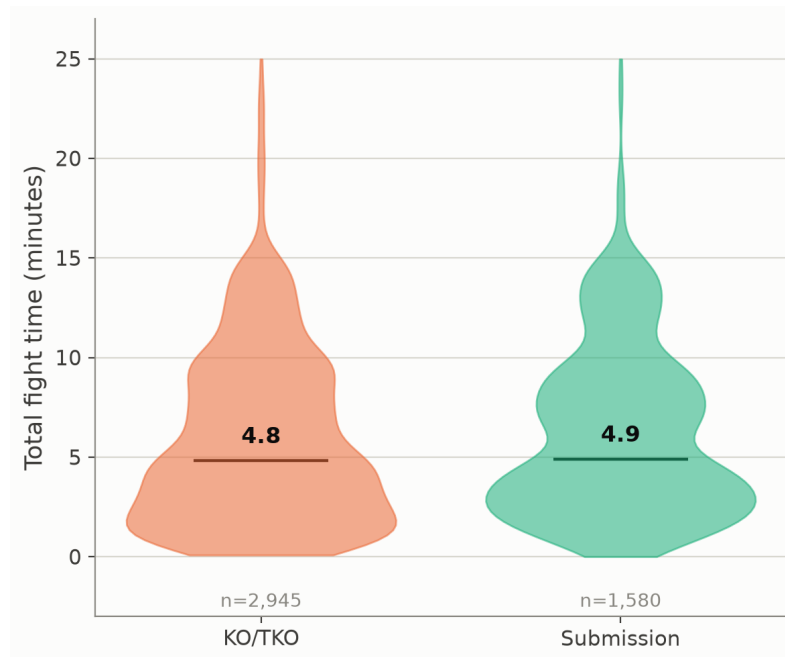


Figure 3: **Most finishes land in round one, no matter the type.** Total fight time for finishes only, split by method.

Both distributions are bottom heavy. The medians for knockouts and submissions are nearly equal, 4.8 minutes against 4.9. Both sit below the first round mark (5 min). It does not matter which type of finish is coming, over half of every finish lands in round one.

5.2 Within submissions, has the technique mix changed?

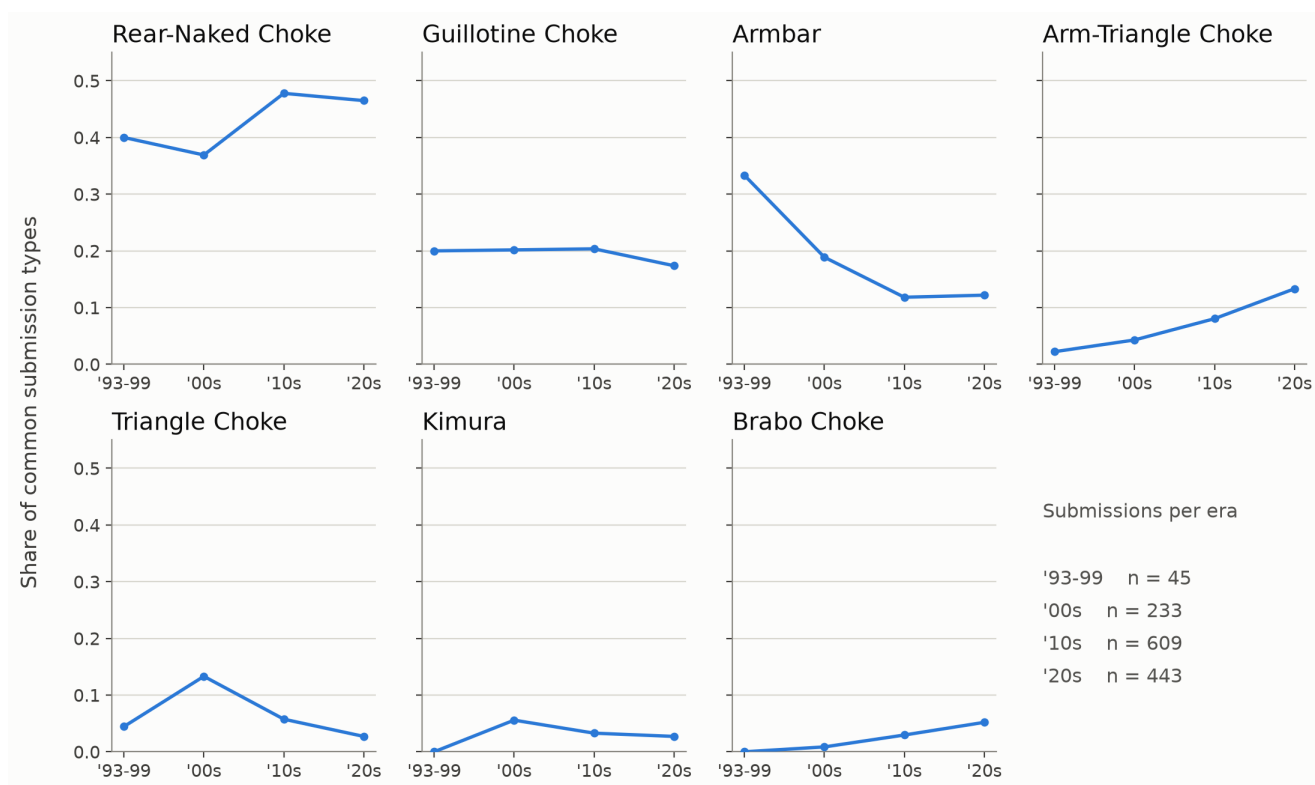


Figure 4: **The arm triangle choke grew as the armbar faded.** The seven most common submission types (84% of all submissions). Counts per era are given in the last panel.

The submission game narrowed around chokes. The rear-naked choke rose from 40.0% to 46.5% of common submissions and the arm-triangle from 2.2% to 13.3%. The armbar collapsed from 33.3% to 12.2%, the largest single movement. Guillotines held roughly flat.

A potential reading is that as takedown and striking defence improved, submissions increasingly arrived from dominant positional control, where chokes are executed, compared to guard positions, where armbars and triangles live.

5.3 Does time off between fights hurt performance?

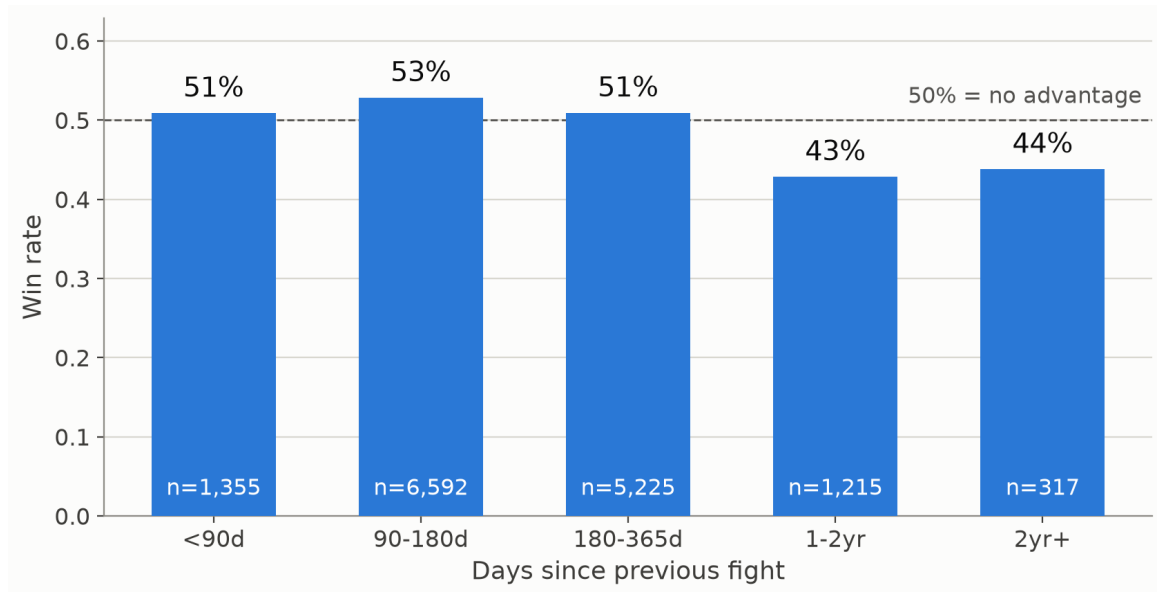


Figure 5: **A short layoff barely matters, but a long one does.** Win rate entering a fight, by days since the fighter’s previous bout.

Win rate is flat at 51–53% for any layoff under a year, then drops to 42.9% at one-to-two years and 43.8% beyond. There is no support for “ring rust” at short timescales, since a fighter returning after ten months does no worse than one returning after two.

Long absences usually coincide with serious injury so this may measure the consequences of whatever forced the layoff rather than the layoff itself.

5.4 When do fighters peak?

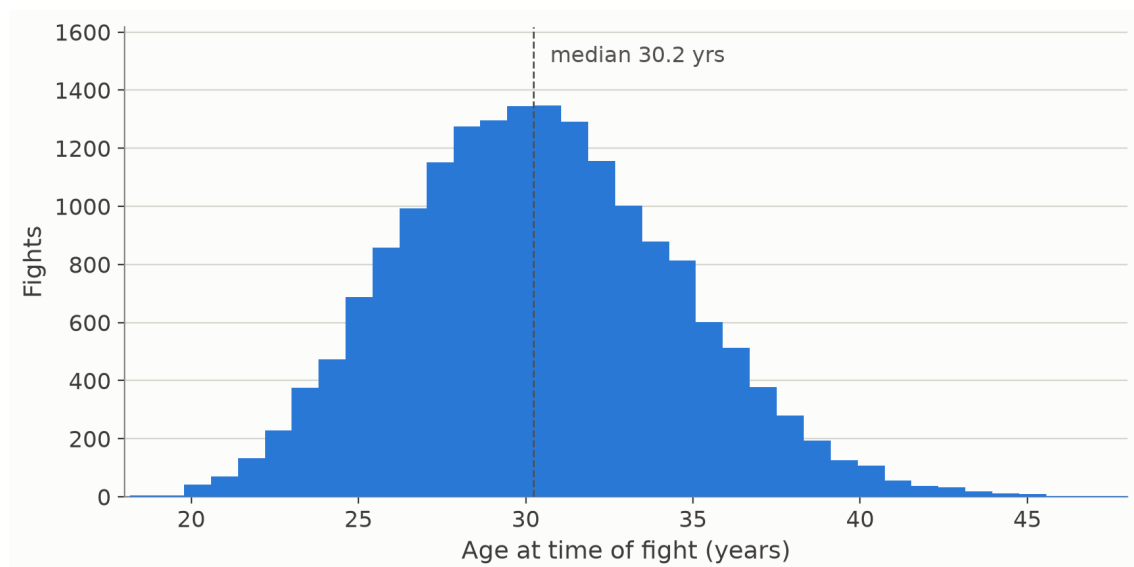


Figure 6: **Almost every UFC fight happens between the ages of 24 and 36.** Distribution of fighter age at the time of each bout ($n = 17,795$ fighter-fights).

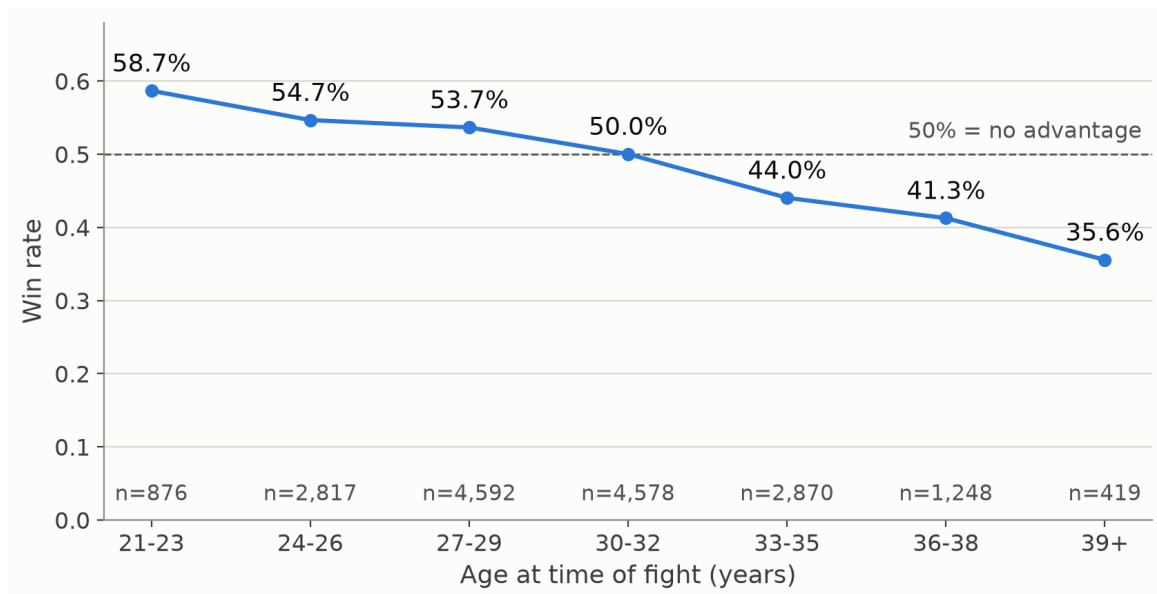


Figure 7: **Win rate falls steadily from a fighter's youngest bouts onward.** Win rate by age bracket at the time of the fight. No physical peak is visible.

Win rate declines across every bracket, from 58.7% at 21–23 to 35.6% at 39 and over, crossing 50% at about age 30. The youngest fighters in the promotion are the most successful.

5.5 Is there a home country advantage?

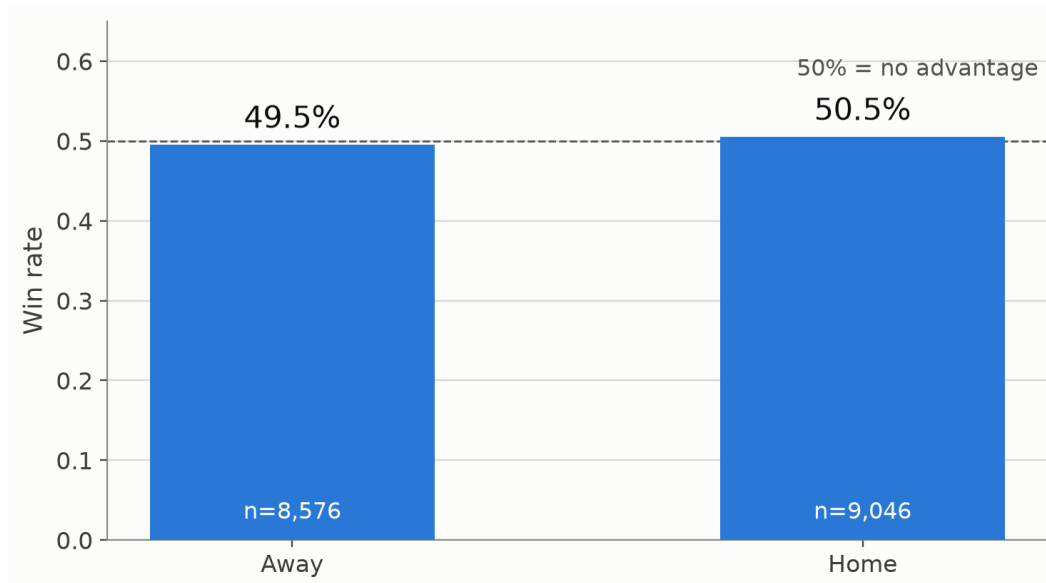


Figure 8: **Fighting at home barely moves the needle.** Win rate for fighters competing in their home country versus abroad.

Fighters in their own country win 50.5% against 49.5% away, one point across 17,622 fighter-fights. Whatever home crowd effect exists is close to negligible in the UFC.

5.6 Do bigger crowds see more finishes?

A second domain, joined onto the first. Wikipedia's attendance figures are matched to Sherdog's events by date.

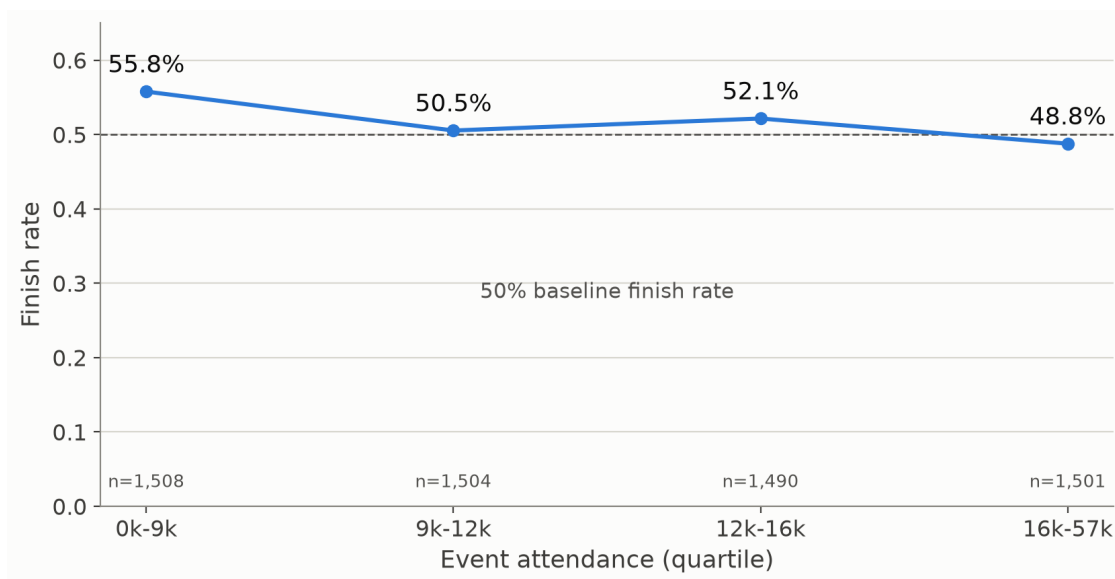


Figure 9: **No evidence that bigger crowds see more finishes.** Finish rate by attendance quartile. If anything the trend moves slightly in the opposite direction.

The finish rate runs 55.8%, 50.5%, 52.1%, 48.8% across the four quartiles, a difference of 7 points between the smallest and largest crowds, and in the opposite direction to what the question asked. In this data the crowd size is not a useful predictor of how a card ends.

5.7 Which active fighters never let a fight reach the judges?

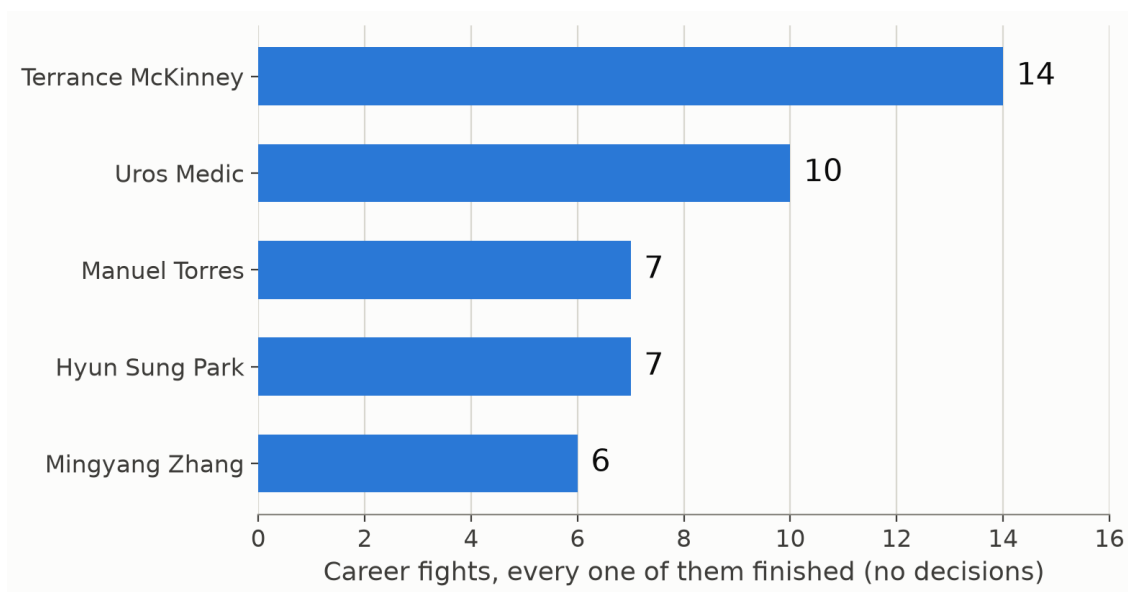


Figure 10: **The longest active perfect-finish streaks.** Active fighters whose every bout has ended in a finish, for or against them. Ranked by streak length. All five have a 100% career finish rate.

Among fighters with at least three bouts in the last eighteen months, five have never had a fight reach the judges. Terrance McKinney leads with fourteen straight finished fights, then Uros Medic with ten.



Figure 11: **What the image pipeline produces.** Two portraits pulled straight out of fighters.csv, with career stats computed from fights.csv alongside the Pillow-derived file metadata. Dricus du Plessis, South Africa’s UFC Middleweight Champion in this dataset, and Terrance McKinney, the streak leader above.

5.8 How accurately can a simple rule call a fight?

Using results and insights uncovered during the analysis of this data, I created a simple signal model to predict the outcome of a fight. Each signal was tested alone first. Three of the chosen signals proved directional and one did not.

Signal	Direction	Evidence
win_rate_entering	higher is better	57.1% alone
age_at_fight	younger is better	\$5.4
days_since_prior	shorter is better	\$5.3
finish_rate_entering	no effect	51.2 / 50.5 / 51.4 / 50.7% by quartile

The finishing rate result is a genuine null. A fighter’s tendency to finish describes more how they win, not whether they will win, so it was dropped after analysis.

Using the other three signals a scorecard is created. Whichever corner is better on a signal scores a point, and the majority of three is the prediction. Ties on all three are recorded as no prediction rather than scored wrong.

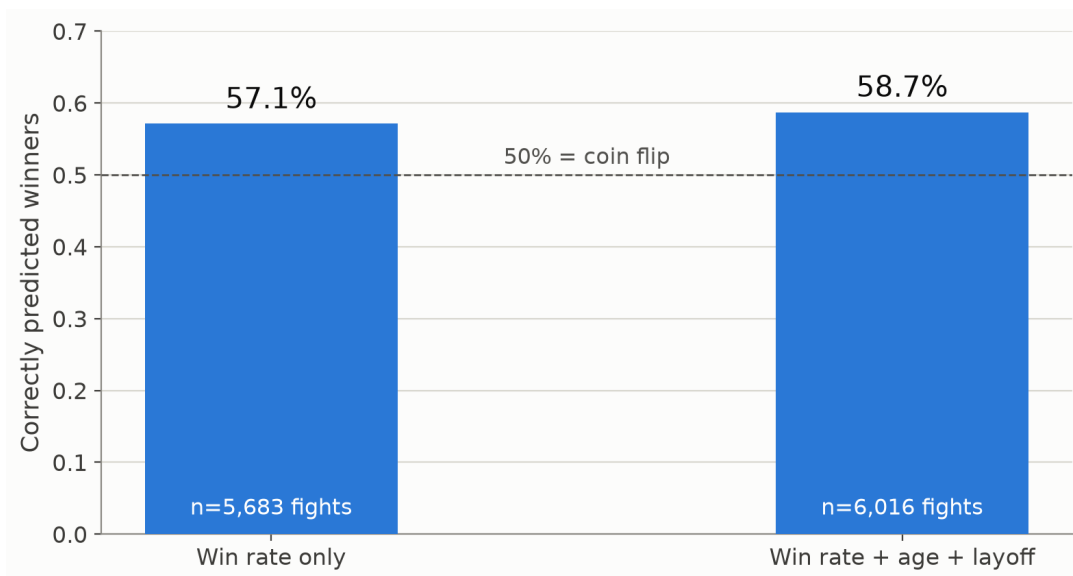


Figure 12: **Three simple pre fight signals get the fight result correct 59% of the time.** Backtested accuracy against a 50% coin-flip reference.

The scorecard calls **58.7%** of 6,016 fights correctly. Career win rate alone calls 57.1% of 5,683, nearly nine points above chance already. Adding age and layoff on top only gains another 1.6 points. The three signals are probably correlated, a fighter with a high win rate also tends to be younger and more active, so they are not adding much independent information.

A correction made during analysis. An early signal version silently gave every exact tie to the same corner. That version measured only 53.4% and made it look like combining signals made the predictor worse, an unexpected result no matter how crude or simple the signal predictor is. After fixing this to favour no fighter during a tie, the accuracy rose to the reported number above. A nice result.

6 Conclusion

The question this report opened with was how much of a fight's outcome is visible before it starts. The answer is some, not much. Three pre-fight signals correctly call 58.7% of fights, a real edge over a coin flip, but most of that edge comes from win rate alone.

The larger pattern sits outside the predictor. The UFC has changed how it ends, decisions have overtaken finishes, and that shift outlasts any single fighter or era. Every timing or crowd effect tested, long layoffs, home advantage, crowd size, came back small or negligible, while what does matter, age and win rate, is about the fighter rather than the occasion.